Quantification: 535_z13c1-2
==========================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	32
interior lacuna count	27
frame-edge lacuna count	5

Interior lacunae only: mean, median and sample SD (n = 27)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2208.67	937.5	2477.05	px^2
major_axis_length_px	lacuna shape	75.7618	65.0635	44.8906	px
minor_axis_length_px	lacuna shape	34.4873	26.9966	19.5676	px
aspect_ratio	lacuna shape	2.2328	2.2373	0.6707	unitless
eccentricity	lacuna shape	0.8529	0.8944	0.1073	unitless
solidity	lacuna shape	0.7855	0.7833	0.0942	unitless
roots_count	headline	6.8519	6	4.6385	count
ring_length_r30_px	headline	218	186	138.252	px
ring_length_r60_px	reported	426.111	389	253.367	px
owned_length_px	ownership-dependent	439.887	340.593	330.175	px
edge_count	ownership-dependent	21.6296	16	18.8723	count
mean_edge_length_px	ownership-dependent	22.3034	20.793	8.6417	px
perimeter_px	normalised	217.567	145.782	172.499	px
ring_area_r30_px2	normalised	7330.37	5469	4110.57	px^2
ring_area_r60_px2	normalised	15324.5	12868	7333.02	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9983	1	0.0069	unitless
ring_density_r30	normalised	0.0296531	0.0290166	0.00962785	px^-1
ring_density_r60	normalised	0.0275734	0.0269705	0.00809862	px^-1
roots_per_100px_perimeter	normalised	3.5754	3.2464	1.8798	per 100 px
ring_attached_length_r30_px	network v2	188.518	157	131.93	px
ring_attached_length_r60_px	network v2	314.296	275	249.809	px
ring_length_w_r30_px	network v2	245.807	213.51	157.452	px
ring_length_w_r60_px	network v2	480.774	439.233	288.331	px
sholl_crossings_r10	network v2	5.4815	5	3.9939	count
sholl_crossings_r20	network v2	5.3333	5	3.2933	count
sholl_crossings_r30	network v2	6.3333	6	3.9321	count
loop_count	loops, report only	0.8148	1	0.8338	count
loop_edge_share	loops, report only	0.1083	0.0968	0.1175	share
ring_width_mean_r30_px	width	7.3132	7.5624	1.1228	px

Per field
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measure	value	units
analysed_area_px2	985102	px^2
total_skeleton_length_px	20589	px
canalicular_length_density_per_px	0.0209004	px^-1
junction_count	377	count
junction_density_per_px2	0.0003827	px^-2
lacuna_count	32	count

lacunae_per_px2	3.248e-05	px^-2
skeleton_component_count	351	count
mean_component_length_px	58.6581	px
median_component_length_px	20	px
field_length_density_w_per_px	0.0235171	px^-1
field_density_without_flagged_per_px	0.0204459	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	815	px
unattached_component_count	49	count
canalicular_length_density_attached_per_px	0.0209004	px^-1
field_length_density_with_unattached_per_px	0.0217277	px^-1
loop_count	26	count
loop_edge_share	0.0762	share
edge_count	1103	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.3811	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	19279	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	520	50.9524	21.26	2.3966	0.9088	0.6061	0.134	15.2558	959.056	True
2	509	42.4483	19.3806	2.1902	0.8897	0.7313	1.0605	62.053	174.511	False
3	3449	94.2445	51.4301	1.8325	0.838	0.8008	-1.3442	87.5868	746.098	False
4	424	32.3309	18.5247	1.7453	0.8196	0.7837	-0.9381	141.42	858.868	False
5	434	32.4903	18.5735	1.7493	0.8205	0.7993	1.0377	151.09	693.419	False
6	418	45.2622	13.5025	3.3521	0.9545	0.8132	-0.8448	294.464	171.199	False
7	429	36.6586	16.6305	2.2043	0.8912	0.8684	0.0347	305.65	1016.41	True
8	1518	84.7212	28.0385	3.0216	0.9436	0.7609	0.4889	372.306	481.817	False
9	424	39.0325	14.1936	2.75	0.9315	0.9002	-0.6027	429.66	215.08	False
10	916	42.5998	32.8395	1.2972	0.637	0.7829	0.4554	434.044	377.415	False
11	6729	150.978	65.1171	2.3186	0.9022	0.7721	-0.3046	496.001	291.95	False
12	3320	105.162	44.1209	2.3835	0.9077	0.7721	-1.4464	464.635	577.918	False
13	3645	102.737	49.6538	2.0691	0.8754	0.8697	0.3597	580.157	513.438	False
14	2801	98.9222	38.3287	2.5809	0.9219	0.8173	0.524	580.027	151.776	False
15	9054	174.453	85.8111	2.033	0.8707	0.6586	0.1605	649.475	679.118	False
16	1209	70.3083	25.9548	2.7089	0.9294	0.8103	-1	737.055	638.021	False
17	3256	105.273	45.222	2.3279	0.903	0.7485	-1.2191	649.232	325.526	False
18	524	31.117	22.6042	1.3766	0.6872	0.8748	-1.3629	760.422	888.611	False
19	597	63.166	16.0382	3.9385	0.9672	0.6958	-1.4137	764.35	853.412	False
20	543	31.4075	23.6295	1.3292	0.6588	0.9282	0.087	780.65	298.357	False
21	455	27.4749	22.5328	1.2193	0.5722	0.9362	1.1384	784.457	318.629	False
22	7705	171.941	75.2669	2.2844	0.8991	0.7155	-1.3369	837.132	645.403	False
23	2929	114.614	39.6482	2.8908	0.9383	0.6816	1.0561	854.515	315.604	False
24	959	66.9611	35.0006	1.9131	0.8525	0.5464	-0.2618	924.19	350.901	False

25	580	52.7819	21.5044	2.4545	0.9132	0.716	1.2958	848.429	862.598	False
26	610	37.8566	20.7792	1.8218	0.8359	0.937	-0.7458	884.313	871.107	False
27	2259	90.1772	47.5611	1.896	0.8496	0.6829	0.6436	985.683	487.498	True
28	6212	244.279	36.0383	6.7783	0.9891	0.7991	-1.5538	1007.31	703.743	True
29								249.851	156.571	False
30								638.193	882.738	False
31								865.038	387.673	False
32								1018.02	884.739	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	7	169	259	182.439	10	18.2439	5.8346
2	6	186	464	511.245	24	21.3019	7.0114
3	9	380	763	1034.62	50	20.6924	7.2345
4	2	249	605	229.853	17	13.5208	6.7267
5	4	140	265	330.492	15	22.0328	5.721
6	5	205	331	109.841	8	13.7301	7.7763
7	3	124	249	165.167	8	20.6458	7.3128
8	7	181	367	123.953	11	11.2685	5.2487
9	6	155	389	597.186	32	18.6621	9.0159
10	11	225	404	769.826	33	23.328	7.6
11	16	453	812	867.252	49	17.699	8.3931
12	7	261	359	225.024	12	18.752	6.6633
13	3	204	417	202.64	7	28.9485	8.5039
14	12	263	531	499.031	24	20.793	7.9127
15	14	552	1120	1268.7	62	20.4629	7.9275
16	5	72	91	211.409	4	52.8523	5.8056
17	13	390	559	403.948	18	22.4416	7.669
18	1	95	271	194.196	8	24.2745	7.5743
19	4	72	221	175.569	11	15.9608	6.8691
20	2	86	206	123.669	7	17.667	8.7709
21	3	75	136	391.836	20	19.5918	8.2757
22	17	514	1013	1148.38	80	14.3547	7.6634
23	12	295	389	520.203	22	23.6456	6.0256
24	8	269	439	340.593	16	21.2871	7.3372
25	4	73	172	269.51	7	38.5014	9.4276
26	1	55	196	13.2426	1	13.2426	4.8535
27	6	270	439	255.208	17	15.0122	8.2074
28	8	283	590	409.534	25	16.3814	8.4722
29	3	174	425	675.357	24	28.1399	7.5624
30	3	121	301	351.037	13	27.0028	6.5929
31	7	141	259	288.338	9	32.0376	7.2929
32	6	115	220	277.693	14	19.8352	

Normalised per lacuna

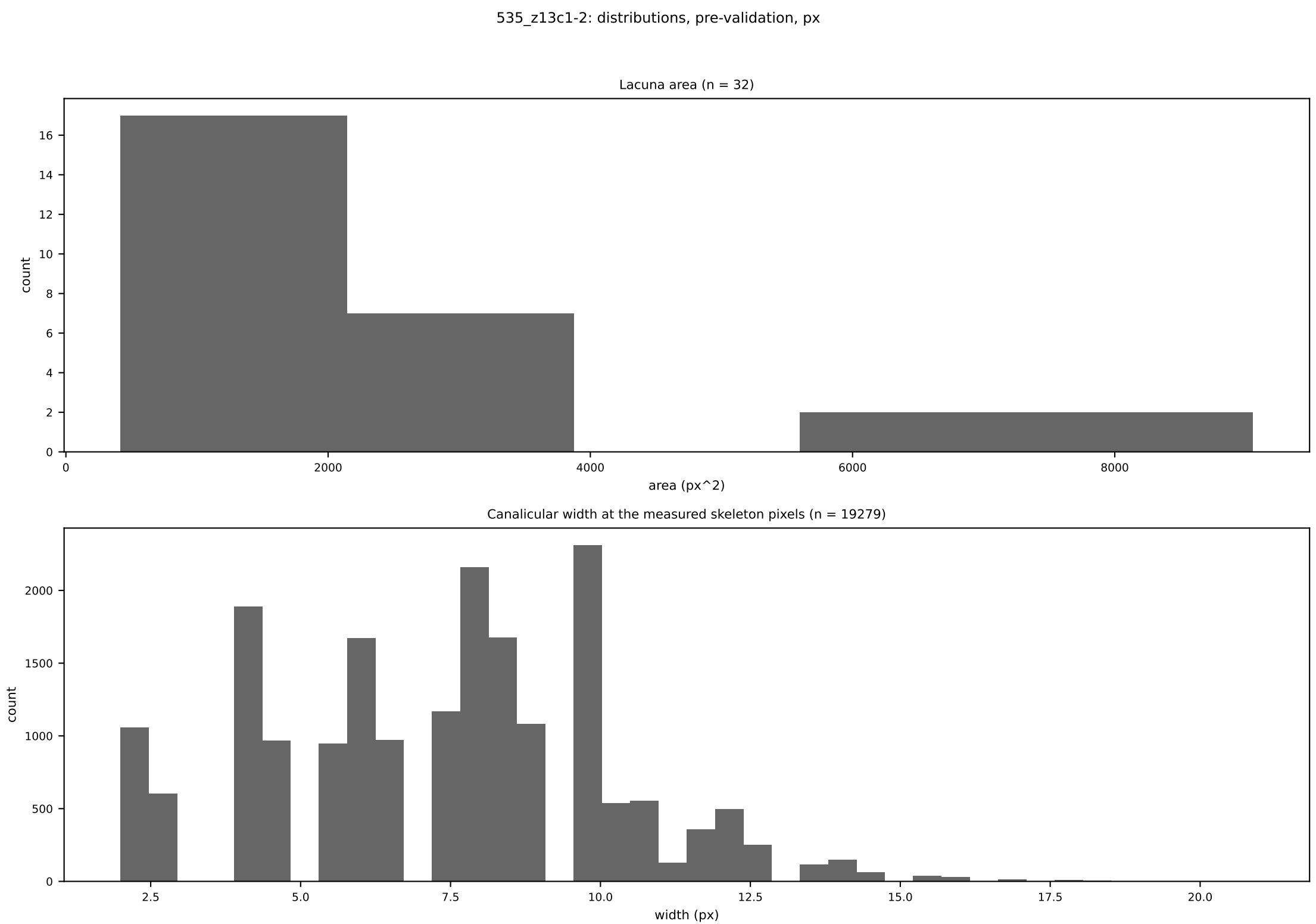
lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	153.297	4649	11757	0.6846	0.6173	0.0363519	0.0220294	4.5663
2	125.882	6125	17204	1	0.9656	0.0303674	0.0269705	4.7664
3	311.848	10440	20760	1	0.9885	0.0363985	0.0367534	2.886
4	101.569	5469	14333	1	1	0.0455294	0.0422103	1.9691
5	104.225	4890	11875	1	1	0.0286299	0.0223158	3.8378
6	104.569	5045	12800	1	1	0.0406343	0.0258594	4.7816
7	96.9056	3199	9195	0.5807	0.5526	0.0387621	0.0270799	3.0958
8	215.622	8573	18773	1	1	0.0211128	0.0195494	3.2464
9	94.5685	5399	12868	1	1	0.028709	0.03023	6.3446

10	145.782	5773	12342	1	1	0.0389745	0.0327337	7.5455
11	433.103	13611	21359	1	1	0.0332819	0.0380168	3.6943
12	326.563	11088	20860	1	1	0.023539	0.01721	2.1435
13	270.936	10229	21506	1	1	0.0199433	0.0193899	1.1073
14	264.007	10205	23680	1	1	0.0257717	0.022424	4.5453
15	731.429	16860	32093	1	1	0.0327402	0.0348986	1.9141
16	194.894	5275	8641	1	1	0.0136493	0.0105312	2.5655
17	328.877	11820	23768	1	1	0.0329949	0.023519	3.9528
18	99.6335	3693	8886	1	1	0.0257243	0.0304974	1.0037
19	147.397	4813	9929	1	1	0.0149595	0.022258	2.7138
20	96.0833	3059	6126	1	1	0.0281138	0.0336272	2.0815
21	85.734	2578	5378	1	1	0.0290923	0.0252882	3.4992
22	655.553	17714	33378	1	1	0.0290166	0.0303493	2.5932
23	380.475	7959	13340	1	1	0.037065	0.0291604	3.154
24	212.208	6331	12629	1	1	0.0424893	0.0347613	3.7699
25	139.498	4945	8638	1	1	0.0147624	0.019912	2.8674
26	100.811	4262	10565	1	1	0.0129047	0.0185518	0.992
27	275.764	8026	15717	0.7864	0.7193	0.0336407	0.0279315	2.1758
28	597.973	10848	20067	0.5965	0.5686	0.0260878	0.0294015	1.3379
29	59.1127	3768	9805	1	1	0.0461783	0.0433452	5.0751
30	65.0122	4634	14443	1	1	0.0261113	0.0208405	4.6145
31	78.9117	3362	7783	1	1	0.0419393	0.0332777	8.8707
32	97.3553	2817	6441	0.5358	0.5236	0.0408236	0.0341562	6.163

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	145	160	197.238	299.806	3	4	2
2	157	275	213.51	530.659	3	4	7
3	323	587	426.848	870.323	10	9	12
4	245	349	279.066	672.487	8	5	7
5	117	210	150.426	294.551	3	6	5
6	145	152	221.953	362.35	5	4	5
7	111	174	143.782	286.978	3	5	4
8	99	121	202.894	411.789	4	6	7
9	128	369	181.681	451.961	3	5	6
10	215	362	253.066	461.789	6	6	6
11	419	763	519.517	925.423	13	9	13
12	189	210	288.309	393.492	6	12	6
13	186	318	226.095	460.948	7	8	5
14	249	430	303.078	599.286	7	8	8
15	528	1065	626.943	1275.47	15	13	16
16	68	87	83.2548	106.397	1	2	1
17	316	361	447.475	633.428	8	8	11
18	83	194	111.054	309.007	3	3	4
19	72	136	77.8701	240.539	2	1	1
20	63	118	94.1838	238.338	2	1	5
21	47	70	80.2843	145.669	1	3	2
22	487	866	576.718	1141.13	15	9	13
23	242	319	333.664	439.233	6	3	4
24	217	320	304.693	490.818	5	7	10
25	73	107	80.6985	193.125	4	2	2
26	14	14	57.799	223.652	0	2	2
27	254	354	304.522	491.718	7	9	7
28	242	346	310.451	653.943	6	10	7
29	172	337	195.782	479.333	5	3	6
30	98	169	137.64	336.907	4	2	4
31	138	177	162.297	292.794	2	3	3

32	108	165	126.154	247.137	2	2	3


535_z13c1-2: distributions, pre-validation, px
Lacuna area (n = 32)
count
area (px^2)
0
2
4
6
8
10
12
14
16
0
2000
4000
6000
8000
Canalicular width at the measured skeleton pixels (n = 19279)
count
width (px)
0
500
1000
1500
2000
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0